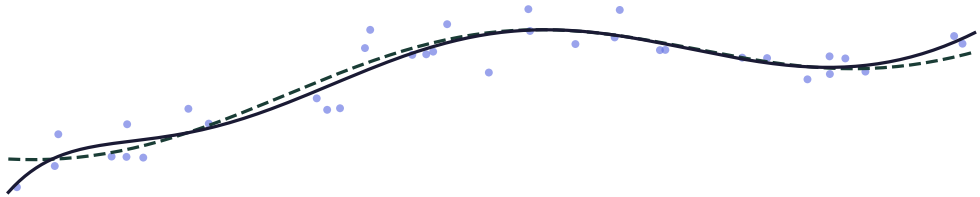
Statistical learning

MATH6331 · CHAPTER 6

Prediction, model flexibility and validation.

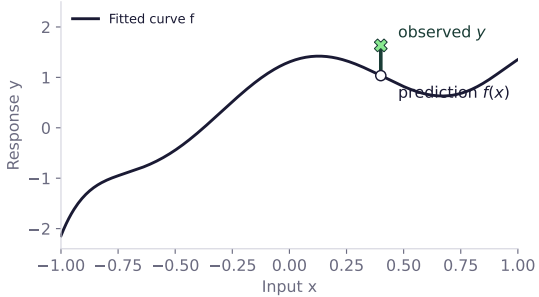


6 November 2026

Error on a new observation

6.2

We have fitted a curve to earlier data. How well does it predict a new response?



One new observation

We observe x and predict $f(x)$. When y becomes available, the squared error is

$$(y - f(x))^2 = (1.63 - 1.03)^2 = 0.36.$$

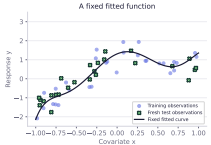
Repeat for other new observations and average their squared errors.

Test loss is the average squared error on a test sample. It estimates **predictive risk** $R(f)$, the average squared error over the population.

Test loss estimates predictive risk

6.2

Fix the fitted function f and population law P, with $R(f) < \infty$. The test observations (X_j, Y_j) are i.i.d. from P, independent of the training data.





Risk bar: numerical integration using the known population law.

Test loss: sample average

$$\widehat{R}_m(f) = \frac{1}{m}\sum_{j=1}^{m}(Y_j - f(X_j))^2.$$

Predictive risk: population average

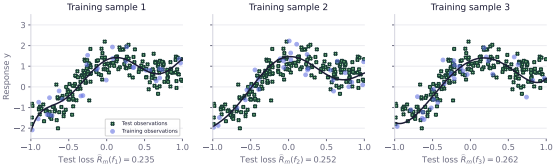
$$R(f) = \int (y - f(x))^2 \, dP(x, y).$$

For this fixed f, $\mathbb{E}[\widehat{R}_m(f)] = R(f)$. Resampling changes the test loss; changing m changes its sampling variability. Neither changes $R(f)$.

Error of a fitting procedure

6.2

Fit the same rule to three training samples: 30 observations, degree 6, least squares.



At most 200 test points shown; all 10,000 used in each test loss.

Assess each curve on the same **10,000 independent test observations**:

$$R(f_i) \approx \widehat{R}_m(f_i) = \frac{1}{m}\sum_{j=1}^{m}(Y_j - f_i(X_j))^2.$$

Their mean test loss is **0.250**. It estimates the risk of the **fitting procedure**.

Why training loss is optimistic

6.2

For a training sample $D = ((X_i, Y_i))_{i=1}^n$ of i.i.d. observations from P, define

$$\widehat{R}_D(f) = \frac{1}{n}\sum_{i=1}^{n}(Y_i - f(X_i))^2.$$

Least squares chooses f_D to minimise this loss over a fixed class $\mathcal{F}$:

$$f_D \in \arg\min_{f\in\mathcal{F}} \widehat{R}_D(f).$$

Assume a minimiser exists and all expectations are finite. Averaging over training samples gives

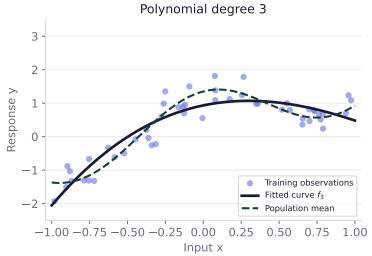
$$\mathbb{E}_D[\widehat{R}_D(f_D)] \le \mathbb{E}_D[R(f_D)].$$

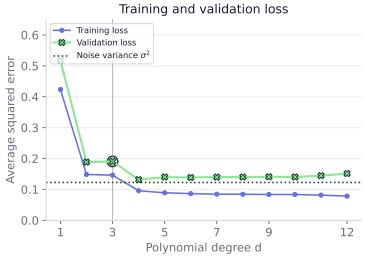
The right-hand side is the risk of the fitting procedure. The inequality need not hold for each individual sample.

Polynomial regression

6.3

For each d, let f_d be the least-squares polynomial of degree at most d.





Dotted line: noise variance, a lower bound for predictive risk only.

Validation loss is the average squared error on 1,200 independent observations. These observations are used to compare degrees, so we call them validation data.

Changing degree keeps both samples fixed. At most 200 validation points shown; all 1,200 used.

Increasing the polynomial degree

6.3

Keep the training sample D fixed. Write

$$\mathcal{P}_d = \text{span}\{1, x, \ldots, x^d\},$$

and let f_d minimise training loss over $\mathcal{P}_d$, as on the previous slide. Since $\mathcal{P}_d \subseteq \mathcal{P}_{d+1}$, the old fit f_d is still available in the larger space. Hence

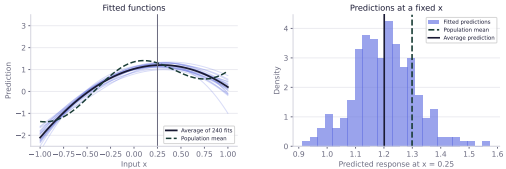
$$\widehat{R}_D(f_{d+1}) = \min_{f \in \mathcal{P}_{d+1}} \widehat{R}_D(f) \leq \widehat{R}_D(f_d).$$

Training loss cannot increase. This gives no ordering of the predictive risks $R(f_{d+1})$ and $R(f_d)$, or of their validation losses.

Repeated training samples

6.4

240 independent training samples; calculations use all fits. Move x to inspect predictions; Visible fits changes only the drawing.



At the selected x, write $z_b = f_b(x)$ and $\bar{z} = B^{-1}\sum_{b=1}^{B} z_b$, with $B = 240$. A fresh response is $Y = \mu + \varepsilon$, where μ is the population mean and $\mathbb{E}[\varepsilon] = 0$.

Squared bias: 0.010

$$(\bar{z} - \mu)^2$$

Variance: 0.012

$$\frac{1}{B}\sum_{b=1}^{B}(z_b - \bar{z})^2$$

Noise variance: 0.122

$$\sigma^2 = \mathbb{E}[\varepsilon^2]$$

Prediction error at x: 0.144

$$\frac{1}{B}\sum_{b=1}^{B}(z_b - \mu)^2 + \sigma^2$$

Why the numbers on slide 8 add up

6.4

Keep the same $B = 240$ predictions z_b at one fixed x. Their average is $\bar{z}$; the population mean response is μ.

Each prediction differs from μ in two parts:

$$z_b - \mu = \underbrace{(z_b - \bar{z})}_{\text{deviation from average}} + \underbrace{(\bar{z} - \mu)}_{\text{error of average}} .$$

The deviations sum to zero: $\sum_b (z_b - \bar{z}) = 0$. So when we square and average, the mixed term vanishes:

$$\frac{1}{B}\sum_{b=1}^{B}(z_b - \mu)^2 = \underbrace{\frac{1}{B}\sum_{b=1}^{B}(z_b - \bar{z})^2}_{\text{variance}} + \underbrace{(\bar{z} - \mu)^2}_{\text{squared bias}} .$$

A fresh response is $Y = \mu + \varepsilon$. Independent noise with mean zero adds its variance σ^2 to this squared error. Thus **prediction error = variance + squared bias + noise variance**.

Choosing and assessing a degree

6.5

Suppose we have 100 observations for choosing a polynomial degree, plus a separate test sample.

Fit on 80 observations. For each $d = 1, \ldots, 8$, fit a degree-d polynomial f_d to these training observations.

Compare on the remaining 20. Compute the validation loss

$$V(d) = \frac{1}{20} \sum_{j=1}^{20} (y_j - f_d(x_j))^2,$$

where (x_j, y_j) are the 20 observations left out of fitting. Choose the degree with the smallest $V(d)$.

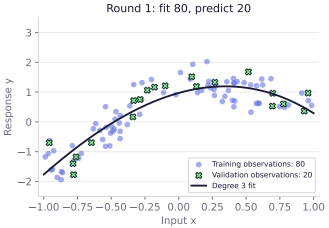
Assess the choice on fresh data. Fit the chosen degree again using all 100 observations. Its test loss is the average squared error on the separate test sample.

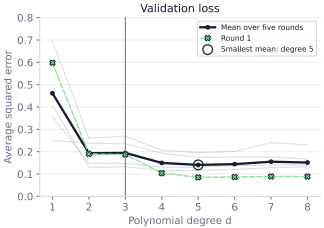
Use each group for validation once

6.5

Split 100 observations into five groups of 20. Each round fits four groups from scratch and predicts the fifth. **Two rounds share 60 training observations, so their errors are not independent.**

New groups keeps the observations; New training sample replaces them. The sliders only inspect existing fits.





For degree **3**: round 1 loss **0.187**; mean over five rounds **0.195**.

For these observations and groups, the smallest mean selects degree **5**. Refit on all **100** observations. Its test loss on **2,000 fresh observations** is **0.168**.

What cross-validation averages

6.5

For each observation i, let $\widehat{y}_i^{(d)}$ be the prediction from the degree-d polynomial fitted **without its group**. Its squared error contributes once:

$$\mathrm{CV}(d) = \frac{1}{100} \sum_{i=1}^{100} (y_i - \widehat{y}_i^{(d)})^2.$$

Each group contains 20 observations, so this is also the mean of the five round losses:

$$\mathrm{CV}(d) = \frac{1}{5} \big(V_1(d) + \cdots + V_5(d) \big).$$

Choose the degree with the smallest $\mathrm{CV}(d)$. Each error above comes from a fit on 80 observations; the final refit uses all 100. Its predictive risk is assessed by the separate test sample.

Questions

6.2-6.5

1 **Increasing degree lowers training loss. Must predictive risk fall?** No. The larger space minimises loss on the training sample, not on the population.

2 **The histogram is wide, but its mean is close to the population mean. Which term is large?** Variance. Squared bias is small because the two means are close.

3 **Why does the smallest CV loss not replace the final test loss?** CV selected the degree and assessed fits on 80 observations. The independent test sample assesses the selected degree refitted on all 100.

Summary

6.2-6.5

Predictive risk is a population average. **Test loss** estimates it from fresh observations.

Increasing polynomial degree cannot increase **training loss**, but may increase **predictive risk**.

At a fixed input x, averaging over training samples and fresh responses gives

$$\text{prediction error} = \text{squared bias} + \text{variance} + \text{noise variance}.$$

Use **validation or cross-validation** to choose the degree. Refit the chosen degree, then assess it on an **independent test sample**.